Hubness, Not Misalignment: A Causal Dial for Cross-Lingual Word Retrieval in Multilingual Models

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Abstract

Multilingual language models reliably translate full sentences across languages; the same cannot be said for single words. This gap is often attributed to word representations being only partially aligned. We show that this failure stems largely from *hubness*, wherein a handful of words (hubs) become the nearest neighbor for far too many translation queries, stranding the rest of the vocabulary (anti-hubs). Via bilingual lexicon induction across a set of eleven diverse languages, we find a few words to be the nearest neighbors of up to 134 queries each, while 82% of the target vocabulary remains stranded; this imbalance and the large hub mass gradually dissolve as the amount of context grows and lexical ambiguity disappears. We find this hubness to be carried by ten high-variance directions of the single-token activation space, emerging as a *bidirectional causal dial*. Projecting out these directions recovers otherwise stranded words and amplifying them worsens retrieval, while a *random* placebo does neither. Additionally, on downstream sentence-level tasks we discover that "de-hubbing" has a greater effect on the hub-dominated XLM-R than on sparsely-hubbed mBERT, and that the hub subspace is present beyond encoder models (a frozen Llama-3-8B). We further reveal that common retrieval-time fixes like CSLS indirectly flatten the same in-degree distribution targeted by our ablation: what looks like misaligned representation can instead be a dial we turn directly within the model geometry.[1]

1 Introduction

Multilingual language models exhibit asymmetrical translation accuracy for different query granularities. On encoders, we measure cross-lingual *sentence* retrieval via hidden states to be 0.96 to 1.00 P@1 across eleven languages, including for typologically distinct pairs like English–Korean and English–Chinese. However, when we switch our queries from sentences to *word* retrieval, accuracy collapses: roughly 0.2 to 0.6 P@1. With no additional alignment training, near-perfect retrieval for sentences falls to barely mediocre when the queries are instead singular words. The standard diagnosis is a misalignment of languages' word vectors; this reading has motivated a large literature surrounding supervised and unsupervised alignment maps (Mikolov et al., 2013; Lample et al., 2018; Artetxe et al., 2018; Cao et al., 2020). We aim to reframe this diagnosis:

> *Word-level cross-lingual "misalignment" is largely a symptom of the* hubness *within single-token activation space; within that hubness, lexical ambiguity is the strongest per-word predictor of failed retrieval. This failed word-level retrieval is the worst-case endpoint of a smooth curve that scales with context and is carried by a small set of high-variance directions within the activation space that can act as a causal dial: we can remove these directions to recover stranded anti-hubs or amplify them to collapse retrieval accuracy.*

Accordingly, we assert that word-level cross-lingual alignment measured by raw nearest-neighbor retrieval largely reflects a hub artifact (Fig. 1) rather than serving as a true proxy for the underlying alignment. The mechanism identified also accounts for much of why field-standard corrections like CSLS (Lample et al., 2018) are effective.

Contributions. (i) We show that the accuracy of word-level retrieval is the worst-case starting point of a monotonically increasing curve that follows context-length, across eleven typologically diverse languages and two encoders (§4). (ii) We identify

[1] Code is available at https://anonymous.4open.science/r/hub-subspace-10DD.

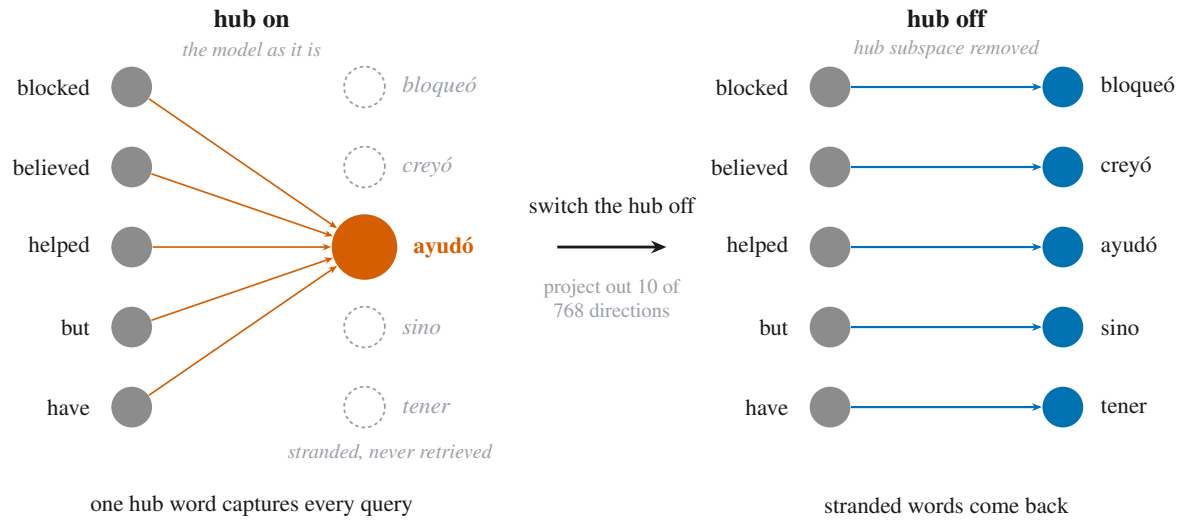


Figure 1: **Word translation fails through a hub. The hub can be switched off.** The examples are en–es on mBERT. Left: each English word retrieves its nearest Spanish neighbor from single-token activations. Many queries go to the same hub word. Most true translations stay stranded. Right: we project out the ten highest-variance directions. The stranded words come back.

the bottleneck of word-level retrieval as ambiguity-driven hubness and show that this hubness inversely follows the curve and collapses as context grows (§5). (iii) We show that this hub structure is concentrated in a small set of top-variance directions, and that the subspace it creates can be utilized as a bidirectional causal dial. We demonstrate that an ablation of the subspace recovers anti-hubs, amplification worsens retrieval, and matched random directions do neither across five language pairs and two model architectures (§6). (iv) We find that the recovery achieved through dehubbing is not redundant to that of a learned alignment map (Procrustes), since an orthogonal map is a rotation and cannot remove hubness from the representation (App. B.5). (v) We discover the presence of the same hub subspace in a frozen decoder. Removing the subspace nudges first-token log-prob in the predicted direction without changing the generated word, though the gain does not survive scoring the full word (Fig. 8). (vi) We relate our findings to classic hubness-in-BLI fixes: CSLS and inverted softmax flatten the same in-degree distribution, and the hub mass they correct at retrieval-time lies partly within the localized subspace (§7).

2 Related work

Cross-lingual alignment and BLI. Mapping word embeddings across languages with a linear transform is classical (Mikolov et al., 2013; Smith et al., 2017), and supervised Procrustes and unsupervised self-learning both reach high accuracy on close pairs (Lample et al., 2018; Artetxe et al., 2018). The difficulty of word-level transfer is documented descriptively (Vulić et al., 2019; Kohli et al., 2024), and Søgaard et al. (2018) bound what any linear map can do by showing the spaces are not fully isomorphic. Feng et al. (2025) find word-level hidden states in decoder LLMs approximately isomorphic and alignable by one orthogonal map, consistent with our account: once a map is fit the representations align, and what raw nearest-neighbor retrieval exposes is hubness. Kargaran et al. (2024) score an English-centric model's per-language competence from parallel-sentence alignment; our granularity result qualifies such scores, reliable at the sentence level but hub-confounded at the single-token level.

The hubness-in-BLI lineage. Hubness, where a few points are the nearest neighbor of disproportionately many queries, was characterized by Radovanović et al. (2010), who tied it to distance concentration in high dimensions rather than to any property of the data. Recent work finds hubness in autoregressive LLMs, where the prediction hubs are benign context-informed frequent tokens (Nielsen et al., 2025). The cross-lingual word hubs we study differ on both counts: they are driven by lexical ambiguity, not frequency (§5), and they are harmful, stranding the words they crowd out. The

BLI literature fixed the problem at *retrieval* time: globally-corrected retrieval (Dinu et al., 2015), the inverted softmax (Smith et al., 2017), CSLS (Lample et al., 2018), hubless nearest-neighbor search (Huang et al., 2019) and the margin criteria of bitext mining (Artetxe and Schwenk, 2019) all penalize promiscuous targets when the similarity matrix is read out. These methods share a stance: the hubness is a nuisance of the similarity matrix, to be corrected when you read it out. None of them localize hubness to activation-space directions, ablate those directions against a random control, or tie word-level hubness to single-token ambiguity; that triad is our contribution. We further show (§7) that these fixes flatten the same in-degree distribution our ablation targets, which gives a representational account of why they work.

Language-identity subspace removal. A separate line projects a small SVD or top-PC subspace out of multilingual encoders to improve cross-lingual sentence and QA retrieval, reading the removed structure as language identity (Libovický et al., 2020; Yang et al., 2021; Xie et al., 2022; Hämmerl et al., 2023). Our subspace is a different object: per-language centering already removes that structure, a held-out language probe sits near chance at 0.38–0.46, yet the hub subspace stays causally active. None of these works measure hubness, and all act at the sentence level rather than the word level (App. J).

Representation geometry and anisotropy. Contextual representations are anisotropic and concentrate variance in a few dominant directions (Ethayarajh, 2019; Timkey and van Schijndel, 2021; Rudman et al., 2022), and removing the top components post-hoc improves embeddings (Mu et al., 2018). Anisotropy is close to hubness, since a shared dominant direction pulls many points toward the same few neighbors, but a space can be strongly anisotropic without any single coordinate being an outlier. Rajaee and Pilehvar (2022) make this point for the multilingual case: the mBERT embedding space is highly anisotropic yet, unlike monolingual English models, shows no rogue outlier dimensions. The hub structure we localize is a high-variance *subspace* rather than one or two interpretable coordinates; ablating dedicated tokenization axes does not de-hub, and those axes are nearly orthogonal to it (App. I). We differ from this line in method and aim. We move from a descriptive geometry statistic to a causal intervention with a downstream behavioral readout (Geiger et al., 2021; Vig et al., 2020), and we tie the geometry to a concrete cross-lingual behavior and its context-length curve. We stay orthogonal to the linear-representation-hypothesis debate (Park et al., 2024), since our directions encode hub structure rather than posited concepts. Our single-phenomenon, layer-aware treatment of multilingual geometry follows the style of Wendler et al. (2024).

3 Setup

Task. In BLI, a source word is matched to its translation in the target vocabulary (the *gallery*, in retrieval terms) by nearest neighbor in a shared activation space; P@1 is top-1 accuracy against a gold dictionary (MUSE en–{es,de,fr,it,zh} (Lample et al., 2018)). For the causal analysis we use *identity* retrieval, raw cosine nearest neighbor in centered activations with no learned map, so that effects are attributable to representation geometry rather than to a probe. Parallel sentences come from FLORES-200 (NLLB Team, 2022), which lets us run the length curve in any language.

Models. We study two masked encoders, mBERT (Devlin et al., 2019) and XLM-R_{base} (Conneau et al., 2020), and a decoder, Llama-3-8B at 4-bit (Grattafiori et al., 2024). Layer-ℓ refers to the hidden-state activations at layer ℓ; words are embedded in isolation (subword-mean), per-language mean-centered, and unit-normalized; Appendix N consolidates all preprocessing in one place. We report encoder layer 1 for word-level retrieval, an early layer where in-degree skew is still mild and word-level BLI sits within about a point of its mBERT peak (§5, App. E). Llama layer 16.

Hubness (Radovanović et al., 2010) is the property that a few hub points are the neighbor of many queries while anti-hubs are no one's neighbor. We use the in-degree N_1, each target's k-occurrence (Radovanović et al., 2010) at $k{=}1$ since BLI is 1-NN retrieval, and summarize hubness by its skewness and by the fraction of gallery items never retrieved (the anti-hub rate). Words whose gold target is an anti-hub are *stranded*: baseline retrieval can never return them. Removing hubness from the representation itself is *hubness reduction* (Feldbauer and Flexer, 2019); we call it de-hubbing. CSLS, cross-domain similarity local scaling (Lample et al., 2018), is the field-standard hubness-corrected simi-

larity measure and serves as our strong baseline.

4 Difficulty scales with context length

We truncate FLORES sentence pairs to their first k words and measure bitext P@1 (encoder masked-mean layer 8). Accuracy rises monotonically with k in all eleven languages and both encoders (Fig. 2). Single tokens are the worst case everywhere: mBERT single-token P@1 ranges from 0.022 (en–ar) to 0.135 (en–de) and reaches 0.96 to 1.00 at full sentences; XLM-R behaves the same, from 0.007 (en–ar) to 0.097 (en–de) up to 0.96 to 1.00. Word-level translation is the worst-case point on this curve, not a special failure of cross-lingual geometry.

The distinctiveness confound, ruled out. A skeptic could argue that longer inputs are just more unique, and so easier to retrieve regardless of alignment. We test two length-matched constructions on en–es and en–zh. *Shuffled* keeps the same k words in scrambled order; *random-bag* draws k words from different random sentences. Random-bags have equal length and similar within-language confusability, yet they stay near 0 (App. D), so length and distinctiveness alone do not drive the curve. Shuffled inputs recover most of the gain (0.77 to 0.82 at full length), and coherent order supplies the rest. Much of the help from context is aggregation of the right content tokens, which (§5) de-hubs the representation, with word order a smaller contributor.

The curve is typologically universal. The eleven targets span four scripts and several morphological types, from fusional Romance to agglutinative Finnish, Turkish and Korean, to logographic Chinese. The shape of the curve does not change. What changes is the height of its left end: the single-token P@1 a language starts from tracks how hard its words are to align in isolation, while the right end is near 1.0 for all of them. If specific pairs were poorly aligned, their gap would persist at longer lengths. It does not, so the phenomenon is a property of the single-token regime, not of particular languages.

5 The mechanism: ambiguity-driven hubness

Hubness collapses with context (Fig. 3). At the single-token level the in-degree distribution is extremely skewed: on mBERT en–es the top hub is the neighbor of 134 queries and 82% of the gallery is never retrieved, with skewness between 1.7 and 2.5. At the sentence level the same distribution is nearly uniform, with peak in-degree 2 and nothing left unretrieved. Averaging tokens washes out the hubs. Across context lengths, hubness and accuracy co-vary strongly ($r = -0.72$, App. H); §6 makes the relationship causal by direct intervention on the hub geometry.

Ambiguity predicts failure. Monosemous words retrieve at 0.52, polysemous words (three or more translations) at 0.26; function words trail content words; and anti-hub *targets* are essentially unretrievable, at 0.02 against 0.65 for hub targets. A multivariate logistic regression of per-word success on all factors at once keeps polysemy as the strongest predictor and the only large one (Table 15, App. G); subword fertility, predictive on its own, washes out once polysemy and word length are held fixed, and frequency adds a smaller independent effect, rarer words being slightly harder. Ambiguity, not rarity, is the driver, and it survives the joint controls.

Why ambiguity produces hubs. The link is mechanical. An ambiguous word, whether a short function word, a polysemous content word or one the tokenizer shatters into generic pieces, has an underspecified representation that lands in a dense central region of the space. Central points are close to many others, so they win the nearest-neighbor competition for many queries and become hubs, the spatial-centrality account of hubness (Suzuki et al., 2013; Hara et al., 2016); the words they crowd out become anti-hubs that no query ever selects. Context resolves the ambiguity: once a word is read in a sentence, its representation moves to a more specific location, the central crowd disperses, and the in-degree distribution flattens (Fig. 3). We do not rest this on a mediation model: a per-word indirect effect of ambiguity through the hub geometry is not distinguishable from zero (App. H), so the evidence is the intervention in §6, not a correlational path.

Where the word signal and the hubness sit across depth. A layer-wise sweep (App. E, Fig. 6) shows that on mBERT word-level BLI peaks at an early layer (layer 4) where in-degree skew is mild, then declines as skew rises with depth. XLM-R is hub-dominated at every layer, and its embedding layer is best for word-level BLI while

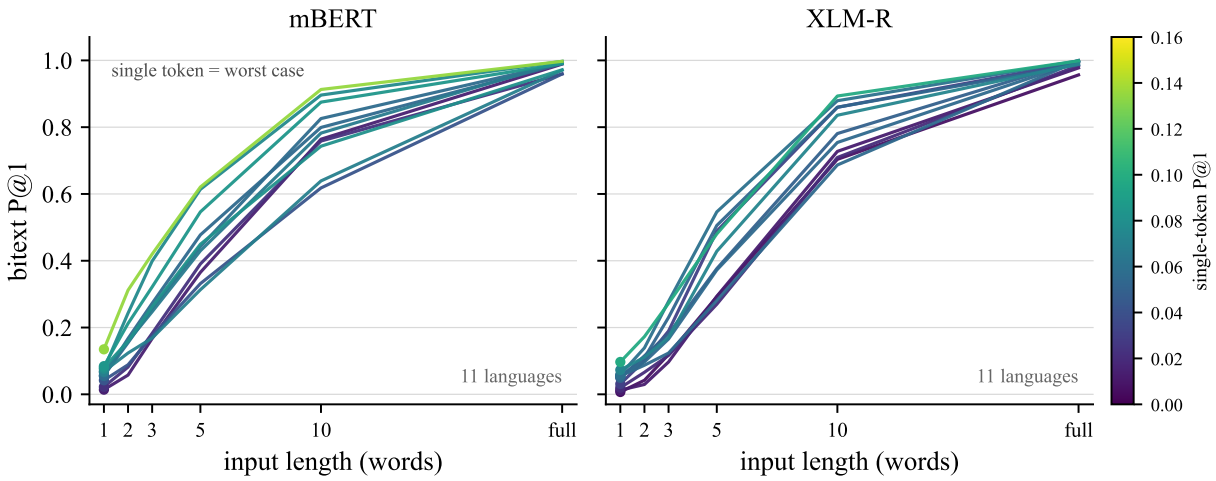


Figure 2: **Difficulty scales with context length in every language.** The plot shows bitext P@1 against input length for eleven languages and two encoders. Color shows single-token P@1. Length 1 is the worst case. The curve rises to near 1.0 at full sentences.

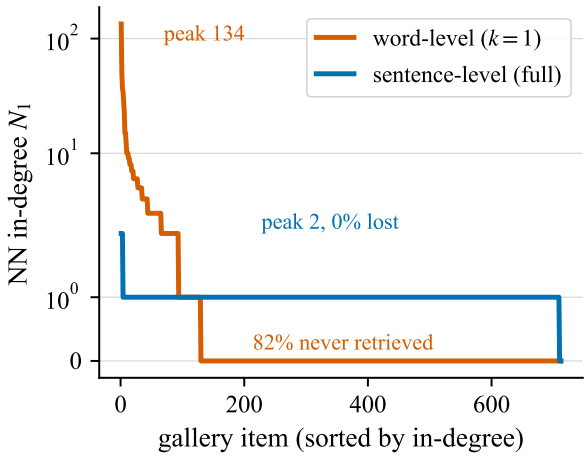


Figure 3: Sorted nearest-neighbor in-degree N_1 on mBERT en–es. The word level gives a large hub spike. The sentence level gives a nearly flat curve.

also the most skewed. We therefore read the cross-layer relation as representation-dependent rather than a universal depth law, and rest on the within-representation claim: at a fixed layer the hub subspace strands anti-hub words, and removing it recovers them.

6 The hub subspace is a causal dial

Causal ablation with a random control (Fig. 4). We take the hub-carrying directions to be the top principal components of the concatenated, per-language-centered bilingual activation space (estimated jointly over the stacked source and target sides), and we ablate them by projecting out the top $k = 3$ to 10 of them from both sides (all-but-the-top, ABTT (Mu et al., 2018)). The spectrum sets k: $k^\star = \max\{k : \lambda_k / \sum_i \lambda_i > \tau/d\}$, every component carrying at least τ times the isotropic share, which at τ=10 selects $k^\star \approx 3$ on mBERT and $k^\star \approx 7$ on XLM-R (App. B.1). These few directions carry a disproportionate share of the variance: the top ten hold about 14% on mBERT and 23% on XLM-R, roughly 11 and 18 times their isotropic share, while the space as a whole stays high-dimensional (App. C). The prediction is specific: this should de-hub and recover anti-hub words while ablating k *random* directions should not. It holds on all five MUSE pairs and both encoders. On mBERT, anti-hub recovery rises from 0.07–0.12 at baseline to 0.14–0.19 under hub-ablation, matching CSLS; on XLM-R, where hubness is more severe, it rises from 0.05–0.09 to 0.18–0.22 and *exceeds* CSLS (0.11–0.13). Random ablation of equal rank stays at baseline everywhere. The random control is what makes the effect attributable to the high-variance hub subspace rather than to generic rank reduction.

The subspace transfers across pairs and word splits. Estimated on en–es alone and applied unchanged, the projection recovers anti-hub words on the four other pairs at 113% of the pair-native gain on mBERT and 93% on XLM-R, while random directions do nothing. A subspace estimated on a disjoint half of the words de-hubs the held-out half as well as the in-sample fit, and the leakage-free strict subset holds (App. B.4, Tables 6–7). The transferred and native subspaces sit 35.0°/30.2° apart, against 84.8° for random ones. Nuisance axes do not carry it: subword-fertility and length directions are near-orthogonal (81–86°), and per-

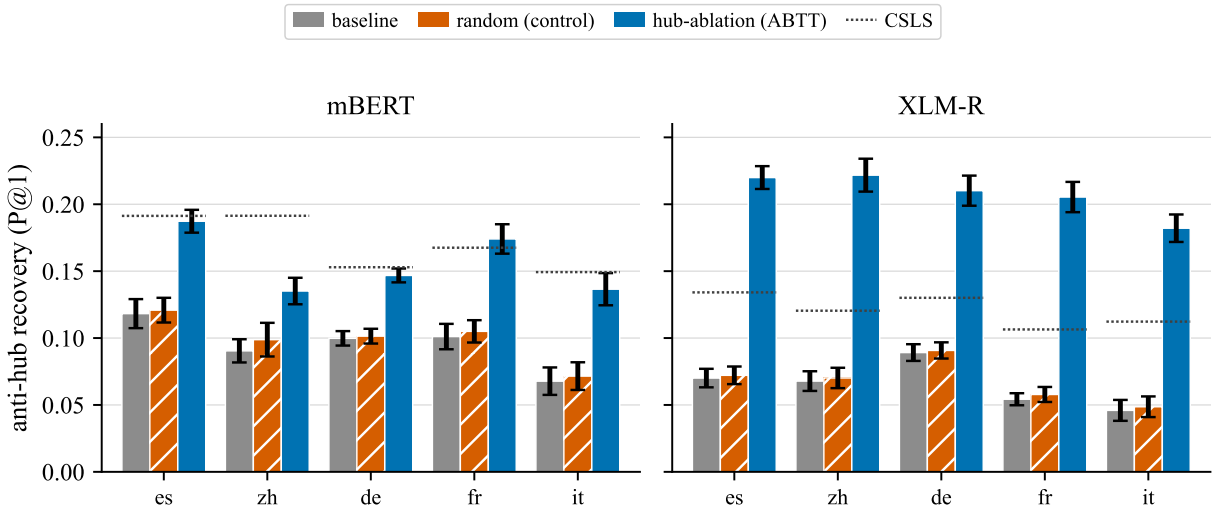


Figure 4: **Causal de-hubbing ablation for five pairs and two encoders.** The bars show anti-hub recovery under three conditions: baseline, random-direction ablation (hatched) and hub-subspace ablation (ABTT). The dotted line shows CSLS. Error bars show 95% CIs over 5 seeds.

language centering leaves no linear language-ID axis (0.38–0.46, near chance), so the transferred object is not the language-identity subspace of LIR- and LSAR-style projections (Yang et al., 2021; Xie et al., 2022) (App. I, J).

The subspace is sufficient to collapse retrieval. Scaling the same top-k subspace *up* ($\alpha > 1$) rather than projecting it out drives P@1 down monotonically and anti-hub recovery toward zero on all five pairs and both encoders (for instance mBERT en–es P@1 0.61 $\rightarrow$ 0.21 at α=8), while amplifying an equal number of random directions stays near baseline (App. M). The hub subspace relieves the failure when removed and reproduces it when amplified, which leaves generic rank reduction unable to account for the ablation effect.

Whitening corroborates the localization. ZCA whitening, fit on the same bilingual space, de-hubs at least as well as ABTT and edges it on the heavy-hub XLM-R (App. B.3): convergent evidence, since two operations on the same top-variance directions give the same recovery while random ablation gives none.

On the severe-hub XLM-R repairing the representation beats reweighting the similarity matrix, because every later comparison inherits the repaired geometry. Relative gains over baseline run $+37\%$ to over $+200\%$ across the five pairs (App. B).

The gain survives a learned map. The causal analysis so far uses map-free identity retrieval, so a fair question is whether de-hubbing still helps once a supervised alignment map is fit. It does. We fit an orthogonal Procrustes map (Smith et al., 2017) on a disjoint training half and score on the held-out half, with centering, the subspace and the map all estimated on the training half (App. B.5). Projecting the hub subspace out before fitting the map beats the matched random-direction control on every cell but one, and the effect repeats the encoder asymmetry of the dial. On the mild-hub mBERT the gain is small (paired ΔP@1 $+0.010$ to $+0.016$, 95% CIs excluding zero on four pairs; en–zh is the single k=10 over-ablation cell), and on the hub-dominated XLM-R it is large (paired ΔP@1 $+0.07$ to $+0.09$ and anti-hub recovery $+0.09$ to $+0.15$, CIs excluding zero on all five pairs; Table 8). An orthogonal map is a rotation, so it cannot remove hubness from the representation; the random control shows that dropping ten arbitrary directions before the map does nothing. De-hubbing composes with a learned map and stacks on retrieval-time CSLS as well.

A frozen-decoder test (Fig. 8). As a scoped check that the subspace is a property of the representation and not of encoder retrieval, we ask whether a frozen decoder carries it too. We test few-shot en$\rightarrow$es word translation by Llama-3-8B generation, with the hub subspace estimated from Llama's own layer-16 activations, not transferred from the encoders: the top ten principal components of the unit-normalized, mean-centered matrix that stacks its last-token en and es MUSE-word

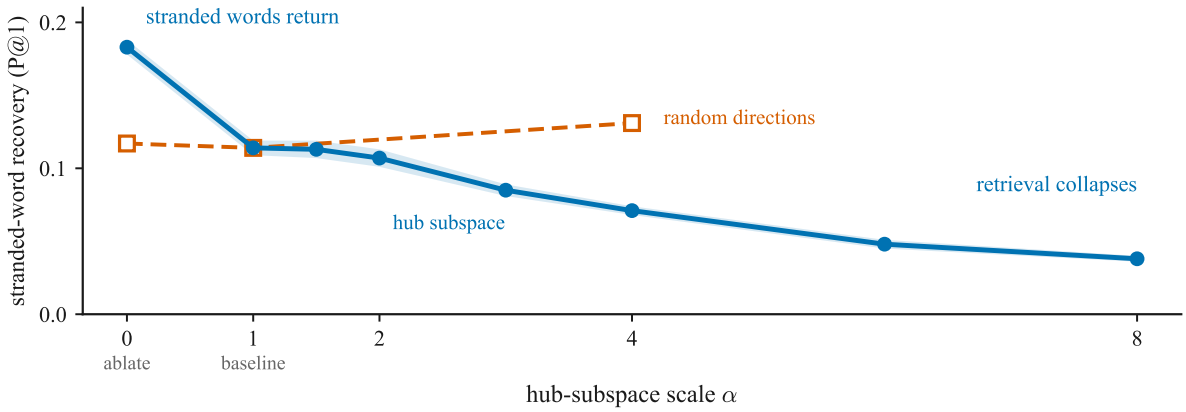


Figure 5: **The hub subspace is a causal dial.** We scale the top-10 high-variance directions by α and measure recovery of stranded words. The model is mBERT at layer 1, en–es. The dashed line scales a matched number of random directions. Points are means over 20 seeds. The band shows the 95% CI.

representations (App. N). Normalization is not optional here. On raw activations the top components are the decoder's few massive activations (Sun et al., 2024), and projecting those out collapses generation (top-1 $0.46 \rightarrow 0.02$), which is a property of those dimensions, not of the hub structure. On the normalized hub subspace the effect is small and one-directional: over 8 seeds and $N{=}150$ test words, projecting it out at layer 16 mid-forward raises the gold word's *first* token's log-probability by a small but reliable amount (paired hub$-$random $\Delta \log p = +0.09 \pm 0.05$ nats, mean $\pm$ 95% CI over the 8 seeds, positive in $8/8$), while first-token top-1 accuracy is unchanged (0.459 to 0.462, $\Delta{+}0.005 \pm 0.013$) and the gold token's median rank stays at 2. The gain does not extend to the whole word: scoring every token of the gold Spanish target teacher-forced, mean log-probability per token moves the other way ($\Delta{-}0.061 \pm 0.041$ nats, negative in $7/8$ seeds), and full-word exact match, every target token the argmax given its gold prefix, is flat (0.297 to 0.298, $\Delta{+}0.001{\pm}0.019$). Random ablation tracks baseline. We read this cautiously. The subspace is present in the decoder's geometry and removing it nudges the first-token distribution the way de-hubbing predicts, but the effect is small, confined to the first token, reverses when the whole word is scored, does not change the generated word, and is sensitive to how the subspace is estimated, so we do not rest a translation-quality claim on it (Sharkey et al., 2025).

A predicted sentence-level effect (App. L). The account also makes a falsifiable prediction one granularity up. Context de-hubs the representation (§4, §5), so single-token hubness should stop mattering once inputs are sentences, *except* where hubness is severe enough to survive contextual averaging. The prediction is therefore directional: removing the hub subspace from frozen cross-encoder sentence features should help zero-shot XNLI transfer on the hub-dominated XLM-R but not on the mild-hub mBERT. It holds. Against the matched random-direction control, ablation raises mean target-language accuracy on XLM-R ($+0.016$, $5/5$ seeds) and does not on mBERT (-0.017, $0/5$), and single-token hubness shows no detectable correlation with the per-language transfer gap ($n{=}14$ languages, all $p \geq 0.43$; App. L), consistent with context already de-hubbing sentence representations. The effect is modest and one-sided, and the paper's headline result stays at the word level. But the sign of a downstream sentence-level intervention matches what the word-level mechanism predicts in advance, on the encoder where it predicts a gain and not on the one where it does not.

7 The retrieval-side fixes correct the same structure

The classic BLI hubness fixes act at retrieval time, on the similarity matrix, not on the representation, a stance that persists for contextual embedding spaces (Nielsen and Hansen, 2023). We ask whether they correct the structure we localize. Two findings (Fig. 7; App. F). First, CSLS and the inverted softmax flatten the same in-degree distribution: on mBERT cosine skew (1.7–2.5) drops under CSLS to 0.9–1.4; on XLM-R, where cosine

skew is severe (9.7–15.8), the inverted softmax flattens it sharply (to 1.9–3.7). Second, the hub mass they correct lives partly in our subspace: the targets CSLS de-hubs have above-chance projection onto the top-k PC subspace (correlation 0.2 to 0.4), and projecting that subspace out (ABTT) reproduces much of CSLS's flattening on its own, with ABTT+CSLS only partly additive. On XLM-R en–zh, ABTT flattens skew more than CSLS does, and ABTT+CSLS still beats either alone. The two interventions therefore share a large common factor, the high-variance hub structure, while each also touches structure the other misses. The inverted-softmax result sharpens this: it barely helps on mBERT, where hubness is mild, but flattens XLM-R dramatically, where hubness is severe, exactly as expected if these fixes act on a hub mass whose size varies by representation. The field's standard retrieval fixes thus have a representational target, and naming it lets one choose where to intervene.

8 Discussion and practical recommendation

The results lead us to the conclusion that word-level cross-lingual representations are not as misaligned as raw nearest-neighbor retrieval would suggest. Instead, much of what may look like misalignment is the hubness tendency of the single-token activation space. And, particularly, this hubness is correlated with lexical ambiguity, dissolves with context and lives within a small high-variance subspace.

Regarding measurement, our practical recommendation follows directly from the conclusion that the granularity of your query may contaminate the reliability of hidden-state retrieval. The nearest-neighbor score for increasingly shorter probes represents hub artifacts more so than misalignment alone. Instead, our recommendation is to measure alignment at a context-length and granularity where hubness is mild, or to correct for hubness specifically with CSLS or inverted softmax. Without these precautions, a study comparing models or layers on raw single-token retrieval via hidden states may mistakenly be ranking them by their tendency towards hubness rather than by the alignment of their single-token spaces.

Some practitioners would rather act on the internal geometric representation of the model than correct for hubness at retrieval-time. For them we demonstrate that our intervention of running a rank-k all-but-the-top projection on an early layer recovers anti-hub words at a rate comparable to CSLS on mBERT and above CSLS on XLM-R. We believe this has utility for use cases where downstream usage consists of a frozen forward pass. Few-shot generation is one such example, where there is no explicit embedding set and therefore no similarity matrix to correct via CSLS or another retrieval-time fix. This intervention is cheap, consisting of a single SVD and projection; runs unsupervised, without the need for parallel data; and can be easily inserted mid-forward into a frozen decoder unlike whitening, as it does not require fitting a full-rank covariance transformation onto the running activations (App. B.3). Our intervention can also be composed with supervised alignment: running Procrustes on previously de-hubbed representations outperforms Procrustes alone on the hub-dominated XLM-R (paired ΔP@1 $+0.07$ to $+0.09$, CIs excluding zero on all five pairs). Additionally, even on the sparsely-hubbed mBERT, initially projecting out the hub subspace in raw representations still leads to marginal gains (paired ΔP@1 $+0.010$ to $+0.016$, 95% CIs excluding zero on four pairs): our interventions can largely stack on existing maps rather than competing with them (App. B.5).

Importantly, we do not propose ABTT as a SOTA BLI method. The value is instead explanatory: while standard metrics treat hubness as a property of the similarity matrix to be corrected at retrieval-time, we show that it is also a localizable and causally active feature of the representation. Accordingly, we urge a reconsideration of measurements and interventions regarding raw nearest-neighbor retrieval on multilingual models.

9 Conclusion

Though word-level cross-lingual translation may be seen as a representation-level failure, we understand it as hubness. Specifically, we pinpoint a phenomenon of many-to-one mapping in single-token activations, corresponding to lexical ambiguity and removed by context. This structure lives in a small and high-variance subspace that can be used as a bidirectional causal dial: ablated to recover stranded words or amplified to strand more, with matched random directions doing neither. Classic retrieval-time fixes correct for structure that overlaps with our subspace (flattening the same in-degree distribution). We also verify this subspace's presence within a frozen decoder. The main

takeaway remains to measure translation accuracy at the right granularity, and to be precautionary about hubness: single-token scores derived from a methodology that does not correct for it are contaminated and not a faithful proxy for cross-lingual alignment. Future work should address whether the subspace found in decoders can support a larger behavioral effect than a shift in first-token log-probability, across different models, directions and layers.

Limitations

Our decoder results are weaker than our encoder results: removing the hub subspace does nudge the gold word's first-token log-prob in the predicted direction ($+0.09 \pm 0.05$ nats, positive in $8/8$ seeds), but it does not move first-token top-1 accuracy (0.459 to 0.462 at $N{=}150$), and the gain does not survive full-word scoring: mean log-probability over all tokens of the gold target moves the other way (-0.061 ± 0.041 nats, $7/8$ seeds) and full-word exact match is unchanged (0.297 to 0.298). The decoder effect is therefore a first-token phenomenon, not a translation-quality effect. Further, this effect is sensitive to the methods used for subspace estimation: utilizing raw activations rather than unit-normalized, the top components turn out to be the decoder's massive activations that dominate raw variance, preventing us from capturing the hub structure. As such, ablating those PCs collapses generation (top-1 0.46 to 0.02) for reasons unrelated to hubness. Therefore, for decoders, we claim that the hub subspace is present in the decoder's geometry but not necessarily that removing it improves translation quality. We also concede that this test is narrow: it covers one decoder, one direction, one layer and one prompt format.

On the encoder side, our directed ablation cannot be considered SOTA in all scenarios. For example, on mBERT, directed ablation matches rather than beats CSLS (en–es 0.628 against 0.636 P@1). Across the ten model–pair cells the ablation closes 8.6% of the word-to-sentence retrieval gap on average (at most 16.0%); the intervention targets anti-hub recovery rather than total retrieval. We instead focus our contribution on its ability to localize the structure and show causality, not the development of a new state-of-the-art retrieval metric. For our mediation claim, we intervene on hub geometry and measure a negligible per-word indirect effect (§6, App. H) rather than fit a statistical mediation model of the length curve (largely because hubness is a global property rather than one held at the word level).

The remaining limitations are further detailed in the appendices.

We are unable to claim a depth law for hubness since mBERT and XLM-R disagree on cross-layer hubness-to-accuracy relationships, so our claim is limited to the subspace at a fixed layer. For sentence-level transfer on hub-dominated XLM-R the direction of the predicted downstream movement holds, though it is one-sided and modest ($+0.016$ mean target-language accuracy, $5/5$ seeds; §6, App. L). This effect is absent in mBERT (-0.017, $0/5$ seeds), where low initial hubness is likely solved for by context alone. This is also in line with existing reports that improvement in encoder alignment does not always buy improved downstream transfer (Veitsman et al., 2026). On the boundary, as we approach genuinely low-resource, different-script language pairs, every method achieves near-zero performance. This is because when the models barely represent the languages, there is no cross-lingual geometry to reliably dehub. Finally, this study was originally motivated by the exploration of relative-representation methods, which did not beat any fair baseline under multi-seed evaluation (App. K).

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A Notation and reporting conventions

All numbers below are recorded experimental outputs; nothing is approximated. Where a value was not measured we say so. Means are over the stated number of seeds, with $\pm$ giving the 95% confidence interval. P@1 is top-1 retrieval accuracy, "max N_1" the peak nearest-neighbor in-degree, "skew" the skewness of the in-degree distribution, "%never" the fraction of gallery items that are no query's top-1, i.e. the anti-hub rate ($N_1{=}0$), and "anti-hub" the recovery rate on words whose gold target was an anti-hub in the baseline condition. What a seed controls depends on the experiment: in the ablation, lineage, amplification, and stability runs it is the resampled $M{=}2000$-word draw and the matched random-direction draw; in the decoder runs it is the $N{=}150$ test-word draw, the in-context example draws, and the random-direction draw; in the XNLI probe it is the 25k-example English training subsample and the random-direction draw. We report per-cell 95% CIs without multiplicity correction; the causal claims rest on the sign being consistent across all model$\times$pair cells and on the matched random-direction controls, not on any single cell.

B Causal ablation across pairs and encoders

Identity retrieval (raw cosine NN in centered layer-1 activations, no learned map), $M{=}2000$ words, 5 seeds. **baseline** is centered only; **CSLS** the hubness-corrected metric; **ABTT-k** projects out the top k principal components (Mu et al., 2018); **random-k** projects out k random directions. The takeaway is identical across both tables: ABTT raises anti-hub recovery well above baseline while random-direction ablation of equal rank stays at baseline; on XLM-R, where hubness is more severe, ABTT also exceeds CSLS. en–zh on mBERT is the one cell where $k{=}10$ over-ablates on P@1 (the subspace is lower-rank there; $k{=}1$–3 are better, see the full per-k runs in the released JSON).

B.1 Sensitivity to the number of ablated directions k

Every ablation seed already sweeps $k \in \{1, 3, 5, 10\}$; Table 3 prints the anti-hub recovery

encoder	pair	baseline	CSLS	ABTT-k10	random-k10
mBERT	en–es	0.118±0.011	0.191±0.012	**0.187±0.009**	0.121±0.009
	en–zh	0.090±0.009	0.191±0.008	**0.135±0.010**	0.099±0.013
	en–de	0.100±0.005	0.153±0.001	**0.147±0.005**	0.101±0.006
	en–fr	0.101±0.009	0.168±0.011	**0.174±0.011**	0.105±0.008
	en–it	0.068±0.007	0.149±0.008	**0.137±0.009**	0.072±0.008
XLM-R	en–es	0.070±0.008	0.134±0.010	**0.220±0.012**	0.072±0.009
	en–zh	0.068±0.007	0.120±0.009	**0.222±0.014**	0.070±0.008
	en–de	0.089±0.006	0.130±0.007	**0.210±0.011**	0.091±0.007
	en–fr	0.054±0.005	0.106±0.006	**0.205±0.012**	0.058±0.006
	en–it	0.046±0.005	0.112±0.007	**0.182±0.010**	0.049±0.006

Table 1: Anti-hub recovery (P@1) by condition, layer 1, 5 seeds. Bold marks the hub-subspace ablation (the paper's intervention), not necessarily the best value: ABTT $\gg$ random $\approx$ baseline in every cell; on XLM-R ABTT also exceeds CSLS, on mBERT it matches CSLS.

encoder	pair	P@1 base→ABTT	skew base→ABTT	%never base→ABTT
mBERT	en–es	0.609 → 0.628	1.74 → 1.19	36 → 34
	en–zh	0.419 → 0.413	2.46 → 2.23	47 → 45
	en–de	0.418 → 0.441	2.44 → 2.01	51 → 48
	en–fr	0.598 → 0.619	1.71 → 1.24	36 → 34
	en–it	0.569 → 0.595	2.18 → 1.49	38 → 36
XLM-R	en–es	0.479 → 0.547	9.68 → 10.02	41 → 37
	en–zh	0.342 → 0.444	15.76 → 5.47	53 → 46
	en–de	0.367 → 0.444	15.32 → 9.64	52 → 45
	en–fr	0.467 → 0.537	7.53 → 5.43	43 → 37
	en–it	0.473 → 0.540	11.97 → 7.15	43 → 37

Table 2: Full-test P@1, in-degree skew, and %never under ABTT-k10 (layer 1, 5 seeds). ABTT lifts P@1 and lowers %never throughout; the skew metric does not always drop on XLM-R (e.g. en–es), where the recovery gain comes from redistributing hub mass rather than flattening it.

at each. The two encoders want different k. On the mild-hub mBERT a small k is enough and a large one starts to cost full-test P@1: en–zh P@1 runs 0.419 (base) $\rightarrow$ 0.443 (k=1) $\rightarrow$ 0.438 (k=3) $\rightarrow$ 0.413 (k=10), the one cell where k=10 over-ablates. On the severe-hub XLM-R recovery climbs monotonically with k (en–es 0.127 $\rightarrow$ 0.220), because the hub subspace is more spread across components there.

A principled, rather than fixed, k follows the spectrum. The top components carry a heavy excess of variance over the isotropic per-dimension share ($1/d$): on mBERT en–es PC_1 holds 4.4% of the variance and the cumulative reaches only 14% by k=10, with the scree flattening (marginal $\approx$ 1% per PC) after about PC_3; on XLM-R en–es PC_1 holds 7.0% and the curve is steeper, 24% by k=10. Choosing k at the scree elbow, where a component's variance falls toward $1/d$, selects a small k on the mild-hub encoder (about 3, beyond which mBERT P@1 degrades) and a larger k on the severe-hub one (still rising at 10), which matches the empirical pattern in Table 3. We report k=10 in the body for a single comparable setting across encoders and note that k=3 is the better default on mBERT. Stated as a rule: with λ_i the eigenvalues of the stacked bilingual matrix and d=768, take $k^\star = \max\{k : \lambda_k / \sum_i \lambda_i > \tau/d\}$, ablating every component that still carries at least τ times the isotropic per-direction share. At τ=10 this gives $k^\star \approx 3$ on mBERT and $k^\star \approx 7$ on XLM-R (leading marginal shares, released with the code).

			ABTT-k			
enc	pair	base	1	3	5	10
mBERT	en–es	0.118	0.169	0.175	0.177	0.187
	en–zh	0.090	0.134	0.143	0.142	0.135
	en–de	0.100	0.126	0.133	0.137	0.147
	en–fr	0.101	0.151	0.158	0.163	0.174
	en–it	0.068	0.105	0.124	0.130	0.137
XLM-R	en–es	0.070	0.127	0.189	0.196	0.220
	en–zh	0.068	0.141	0.174	0.213	0.222
	en–de	0.089	0.120	0.158	0.175	0.210
	en–fr	0.054	0.119	0.169	0.187	0.205
	en–it	0.046	0.092	0.131	0.151	0.182

Table 3: Anti-hub recovery (P@1) by number of ablated PCs k (layer 1, 5 seeds; CIs $\leq$ 0.014, omitted). mBERT saturates by $k\approx$3 (and en–zh full-test P@1 drops past it); XLM-R keeps gaining through k=10.

enc	pair	mean ∠	max ∠	corr	var
mBERT	en–es	13.9±0.5	29.1	0.963	0.141
	en–zh	16.0±1.2	55.5	0.933	0.129
	en–de	15.6±1.6	46.5	0.940	0.140
	en–fr	13.3±0.4	31.1	0.965	0.149
	en–it	16.7±1.1	50.9	0.933	0.139
XLM-R	en–es	9.5±0.5	22.6	0.981	0.239
	en–zh	11.4±1.4	44.4	0.956	0.214
	en–de	8.1±0.3	18.0	0.987	0.235
	en–fr	8.3±0.2	15.9	0.987	0.244
	en–it	9.8±0.6	24.7	0.979	0.226

Table 4: Top-10 PC subspace stability across the 5 seeds (layer 1, M=2000). Principal angles ∠ in degrees; mean and max are over the 10 (non-independent) between-seed subspace pairs, and ± here is the 95% CI over those pairs. "corr" is the mean cosine of the principal angles (1 = identical subspaces); "var" is the fraction of total variance the top-10 subspace carries.

B.2 Stability of the hub subspace across seeds

The hub subspace is estimated from a resampled draw of M=2000 words, so a fair worry is that the top-k directions are sample noise. They are not. We recompute the top-10 PCs of the concatenated bilingual space for each of the 5 seeds and measure the principal angles between every pair of seed subspaces (Table 4). Mean principal angles are 13–17° on mBERT and 8–11° on XLM-R, with mean canonical correlation 0.93–0.99; the more dominant the subspace (it carries more variance on XLM-R), the tighter it is reproduced. The instability sits in the smallest of the 10 directions: the largest principal angle reaches 30–55° on a few mBERT cells, so the bottom of the subspace is the part that moves, not the top.

Stability is sample-size dependent, as expected. On en–es the mean principal angle shrinks from 35° at M=500 to 24° (M=1000), 14° (M=2000), and 0° at M=4000, where all seeds draw the full pool ($XLM-R$: 26° → 19° → 12° → 0°). So M=2000 is already in the stable regime, and the residual angle is a finite-sample effect that vanishes with more words rather than a sign of an ill-defined subspace.

B.3 Whitening versus directed ablation

ABTT removes the top-k subspace outright. A standard isotropy baseline instead rescales all directions to equal variance: ZCA whitening, fit on the same concatenated bilingual space. Comparing the two isolates whether the de-hubbing needs a targeted low-rank projection or just generic anisotropy removal (Table 5). Whitening de-hubs at least as well as ABTT and recovers anti-hub words comparably or slightly better, on both encoders, while discarding no directions. On the mild-hub mBERT the two track each other closely (en–es anti-hub 0.118 base, 0.187 ABTT, 0.182 ZCA; skew 1.74/1.19/1.20). On the severe-hub XLM-R whitening flattens in-degree more than ABTT and edges it on recovery and P@1 (en–es skew 9.68/10.02/5.75, anti-hub 0.070/0.220/0.256, P@1 0.479/0.547/0.570).

This sharpens the localization claim rather than weakening it. The de-hubbing lives in the anisotropic top-variance subspace, the same directions whitening rescales and ABTT removes; that two different operations on those directions both work pins the effect to them. It is not generic rank reduction: the equal-rank random-direction control (Table 1) does nothing. What we do not claim is a unique minimal set of special coordinates. The hub variance concentrates in a few top-variance directions within an otherwise high-dimensional space (App. C), not a handful of named axes, and the interpretable-axis negative (App. I) makes the same point. ABTT keeps one practical advantage where whitening cannot run: it is a fixed low-rank projection that can be inserted mid-forward into a frozen decoder (§6), removing the hub directions in place without estimating and applying a full covariance transform to the running activations. As a representation-space correction on the retrieval side, the two are interchangeable.

B.4 Out-of-sample transfer of the hub subspace

Two controls separate subspace estimation from evaluation (protocol otherwise identical to Table 1: layer 1, M=2000, 5 seeds, identity retrieval). **Cross-pair:** the top-10 subspace is estimated on the en–es draw only and the projection is applied unchanged to the other four pairs; "strict" further restricts scoring to anti-hub queries whose English word never entered the estimation sample (the en sides of the MUSE dictionaries overlap, 34–49% per pair, so the strict column is the leakage-free number). **Cross-split:** within each pair the deduplicated rows are split into disjoint halves; the per-language centering means and the subspace are estimated on the train half only and applied to the held-out half, so no evaluated word in either language touches the estimation. Table 6 and Table 7 give the numbers. Transferred projections recover anti-hub words at 113% (mBERT) and 93% (XLM-

		baseline			ABTT-k10			ZCA whitening		
enc	pair	P@1	skew	anti	P@1	skew	anti	P@1	skew	anti
mBERT	en–es	.609	1.74	.118	.628	1.19	.187	.624	1.20	.182
	en–zh	.419	2.46	.090	.413	2.23	.135	.394	2.26	.126
	en–de	.418	2.44	.100	.441	2.01	.147	.440	1.90	.149
	en–fr	.598	1.71	.101	.619	1.24	.174	.618	1.24	.168
	en–it	.569	2.18	.068	.595	1.49	.137	.592	1.47	.138
XLM-R	en–es	.479	9.68	.070	.547	10.02	.220	.570	5.75	.256
	en–zh	.342	15.76	.068	.444	5.47	.222	.449	6.03	.227
	en–de	.367	15.32	.089	.444	9.64	.210	.459	6.49	.224
	en–fr	.467	7.53	.054	.537	5.43	.205	.555	4.41	.224
	en–it	.473	11.97	.046	.540	7.15	.182	.555	3.91	.207

Table 5: ZCA whitening vs directed top-10 ablation (layer 1, 5 seeds). "anti" is anti-hub recovery. Whitening matches or beats ABTT on de-hubbing (skew) and recovery while discarding no directions; the gap is largest on the heavy-hub XLM-R, where whitening flattens in-degree most.

R) of the pair-native gain, and the train-fit subspace recovers the held-out half within the confidence interval of the in-sample fit on all ten cells, while the random control stays at baseline. Reference-to-native principal angles run 25–49° (mBERT) and 25–35° (XLM-R) against 84.8° for independent random ten-dimensional subspaces, and the five pair-native subspaces sit 25–51° from one another, far from random. The residual angle between estimation contexts is the same order as the between-seed sampling angle of Table 4, so the transferred subspace is the same object up to sampling noise, not a lucky rotation.

B.5 Hub ablation on top of a learned map

The causal dial (§6) is scoped to map-free identity retrieval so that effects are attributable to representation geometry. A BLI reviewer will ask the next question directly: once a supervised alignment map is fit, does de-hubbing the representation still help, or does the map absorb the hubness gain? We answer it on the same ten cells with the same random-direction control.

Protocol. We split each de-duplicated dictionary into disjoint train and test word halves (M=2000 per half where the pair is large enough, 5 seeds, layer 1). Everything is estimated on the train half only: the per-language centering means, the top-10 hub subspace (stacked SVD, as in Table 1), and an orthogonal Procrustes map W (Smith et al., 2017), $W = \arg\min_{W^\top W=I} \|X_{\text{tr}}W - Y_{\text{tr}}\|_F$. We score P@1 and anti-hub recovery on the held-out test half, so no evaluated word in either language touches estimation. Conditions: **identity** (centered, no map), **map** (Procrustes), **ABTT+map** (project the hub subspace out of both sides, then fit and apply the map on the de-hubbed space), **random+map** (the matched-rank random-direction control, same pipeline), and the two CSLS read-outs of the mapped space. Anti-hub queries are defined by baseline (identity) in-degree on the test half, as elsewhere.

The map does not absorb the gain. Table 8 gives P@1 and Table 9 anti-hub recovery. ABTT+map beats random+map on nine of ten cells, and the split follows the encoder asymmetry of the dial. On the mild-hub mBERT the gain is small but consistent (paired ΔP@1 $+0.010$ to $+0.016$ on the four non-zh pairs, 95% CIs excluding zero; en–zh is the one k=10 over-ablation cell noted in Table 3, -0.005 ± 0.007). On the hub-dominated XLM-R it is large (paired ΔP@1 $+0.070$ to $+0.093$, anti-hub recovery $+0.086$ to $+0.145$, CIs excluding zero on all five pairs), where the orthogonal map on its own barely improves over identity (en–es anti-hub 0.072 identity, 0.143 map, 0.288 ABTT+map). The random-direction control sits at the map baseline in every cell (P@1 within 0.002, anti-hub within 0.002), so the gain is specific to the hub subspace, not to removing ten directions. An orthogonal Procrustes map is a rotation and preserves within-language nearest-neighbor structure, so it cannot remove hubness; projecting the hub subspace out first does, and the benefit survives the map. De-hubbing also stacks on the retrieval-time fix: ABTT+map+CSLS is the best condition on nine of ten cells (e.g. XLM-R en–es P@1 0.486 map, 0.503 map+CSLS, 0.577 ABTT+map, 0.595 ABTT+map+CSLS); the exception is mBERT en–zh, the k=10 over-ablation cell, where map+CSLS is higher. The intervention is a lever that composes with a learned map, not a

enc	pair	baseline	random-k10	ABTT native	ABTT transfer	transfer (strict)	$\angle$(ref, native)
mBERT	en–es	0.118±0.011	0.122±0.010	0.187±0.009	0.187±0.009	(ref)	0° by construction
	en–de	0.100±0.005	0.102±0.005	0.147±0.005	0.142±0.004	0.131±0.006	37.2°
	en–fr	0.101±0.009	0.105±0.008	0.174±0.011	0.177±0.007	0.180±0.010	25.1°
	en–it	0.068±0.010	0.071±0.010	0.137±0.012	0.131±0.009	0.122±0.010	29.1°
	en–zh	0.090±0.009	0.099±0.010	0.135±0.010	0.162±0.008	0.155±0.009	48.6°
XLM-R	en–es	0.070±0.007	0.072±0.007	0.220±0.009	0.220±0.009	(ref)	0° by construction
	en–de	0.089±0.006	0.091±0.006	0.210±0.011	0.195±0.012	0.184±0.015	32.4°
	en–fr	0.054±0.004	0.058±0.006	0.205±0.011	0.196±0.008	0.195±0.010	24.8°
	en–it	0.046±0.008	0.049±0.009	0.182±0.010	0.167±0.008	0.166±0.008	28.5°
	en–zh	0.068±0.007	0.070±0.008	0.222±0.012	0.222±0.015	0.208±0.014	35.0°

Table 6: Cross-pair transfer of the hub subspace (anti-hub recovery, layer 1, 5 seeds, ±95% CI). "ABTT transfer" projects out the subspace estimated on en–es only; "strict" scores only queries whose English word was absent from the estimation sample. $\angle$ is the mean principal angle between the transferred and pair-native subspaces (degrees).

enc	pair	baseline	CSLS	random-k10	ABTT train-fit	ABTT in-sample	$\angle$(train, test)
mBERT	en–es	0.115±0.009	0.188±0.007	0.119±0.011	0.185±0.004	0.183±0.007	18.8°
	en–de	0.092±0.008	0.144±0.012	0.093±0.008	0.140±0.012	0.138±0.010	22.3°
	en–fr	0.106±0.013	0.177±0.015	0.111±0.011	0.175±0.013	0.177±0.013	18.7°
	en–it	0.060±0.005	0.131±0.010	0.065±0.003	0.123±0.011	0.126±0.007	23.8°
	en–zh	0.081±0.011	0.179±0.008	0.087±0.009	0.127±0.010	0.126±0.010	22.2°
XLM-R	en–es	0.071±0.005	0.133±0.008	0.075±0.006	0.209±0.011	0.209±0.012	14.6°
	en–de	0.082±0.008	0.122±0.008	0.084±0.007	0.201±0.004	0.204±0.006	10.8°
	en–fr	0.058±0.006	0.105±0.010	0.061±0.007	0.205±0.012	0.204±0.011	12.4°
	en–it	0.042±0.005	0.102±0.007	0.045±0.004	0.169±0.001	0.170±0.005	12.8°
	en–zh	0.054±0.006	0.115±0.006	0.058±0.006	0.211±0.008	0.204±0.006	17.6°

Table 7: Cross-split transfer (anti-hub recovery on the held-out word half, layer 1, 5 seeds). Centering means and subspace are fit on the disjoint train half ("train-fit") or on the evaluated half itself ("in-sample"); no evaluated word touches the train-fit estimation.

substitute the map makes redundant.

C How concentrated is the hub subspace?

We asked whether "low-rank" is the right word. It is not, and the honest answer sharpens the claim. On the concatenated, per-language-centered, unit-normalized bilingual activation matrix, the space the ABTT PCs are taken from (layer 1, M=2000, 5 seeds, d=768), we measure the participation ratio $\mathrm{PR} = (\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$ and the effective rank $\mathrm{erank} = \exp(-\sum_i p_i \log p_i)$, $p_i = \lambda_i / \sum_j \lambda_j$ (Roy and Vetterli, 2007). The representation as a whole is *not* low-dimensional: effective rank is about 405 on mBERT and 300 on XLM-R (Table 10), several hundred of the 768 directions. So "low-rank" overstates it. What concentrates is the *causally-relevant* variance. The top ten principal directions hold about 14% of the variance on mBERT and 23% on XLM-R, roughly 11 and 18 times their isotropic share of $10/768$; the concentration is stronger on the more hub-prone XLM-R, while the rest of the spectrum stays high-dimensional. This concentration, not a small effective rank, is what the causal results act on, and it explains the whitening comparison. The per-k anti-hub recovery in Table 3 tracks the cumulative variance captured at each k (Pearson r=0.77 across the 40 cell-by-k points): on mBERT it saturates by $k\approx3$, but on XLM-R it is still rising at k=10, so the causally-relevant hub structure there reaches past the ten directions a fixed ABTT projection removes. Whitening, which rescales every high-variance direction at once, catches that residual, which is why it matches or edges ABTT on XLM-R.

D Length curve, full numbers

FLORES, encoder masked-mean layer 8, raw cosine retrieval, test block of 712 sentences. Accuracy climbs monotonically with length in all eleven languages and both encoders, and single tokens are the worst case everywhere. Prefixes are cut at word boundaries per language: Chinese is segmented with jieba, since FLORES Chinese is unspaced and whitespace splitting would leave the target side untruncated; the other ten targets are whitespace-delimited.

enc	pair	identity	map	ABTT+map	random+map	Δ(ABTT−rnd)	map+CSLS	ABTT+map+CSLS
mBERT	en–es	0.610	0.650	**0.660**	0.649	+0.010±0.004	0.686	0.689
	en–de	0.414	0.459	**0.469**	0.459	+0.010±0.008	0.486	0.492
	en–fr	0.603	0.637	**0.652**	0.636	+0.016±0.004	0.675	0.680
	en–it	0.566	0.608	**0.621**	0.608	+0.012±0.002	0.647	0.654
	en–zh	0.425	0.406	0.399	0.404	−0.005±0.005	0.451	0.428
XLM-R	en–es	0.474	0.486	**0.577**	0.484	+0.093±0.015	0.503	0.595
	en–zh	0.343	0.338	**0.408**	0.338	+0.070±0.007	0.366	0.426
	en–de	0.367	0.380	**0.460**	0.379	+0.082±0.004	0.402	0.477
	en–fr	0.478	0.486	**0.571**	0.486	+0.086±0.009	0.508	0.592
	en–it	0.468	0.486	**0.566**	0.485	+0.081±0.009	0.515	0.586

Table 8: Hub ablation on top of a learned Procrustes map (full-test P@1, layer 1, 5 seeds, leakage-free train/test split). Bold marks ABTT+map, the paper's intervention (ABTT+map+CSLS is higher still on every cell). Δ is the paired seed-level ABTT+map minus random+map difference, mean ± 95% CI over the 5 seeds; the CI excludes zero on nine of ten cells (all but mBERT en–zh). Per-condition 95% CIs are ≤ 0.022, omitted. random+map tracks map, so representation-space de-hubbing adds on top of the retrieval-time fix.

enc	pair	identity	map	ABTT+map	rnd+map	Δ
mBERT	en–es	0.123	0.253	**0.272**	0.252	+0.021
	en–de	0.098	0.191	**0.207**	0.190	+0.016
	en–fr	0.114	0.223	**0.256**	0.222	+0.034
	en–it	0.061	0.191	**0.216**	0.191	+0.024
	en–zh	0.087	0.197	0.198	0.197	+0.001
XLM-R	en–es	0.072	0.143	**0.288**	0.143	+0.145
	en–zh	0.067	0.135	**0.221**	0.134	+0.086
	en–de	0.088	0.136	**0.236**	0.135	+0.100
	en–fr	0.070	0.126	**0.265**	0.125	+0.140
	en–it	0.045	0.119	**0.245**	0.118	+0.128

Table 9: Anti-hub recovery for the same conditions (test half, layer 1, 5 seeds). Bold marks ABTT+map, the paper's intervention. Δ is the paired ABTT+map minus random+map gain (mean over 5 seeds); the paired 95% CIs are ≤ 0.024 and exclude zero on every cell except mBERT en–zh. The learned map lifts anti-hub recovery over identity, and de-hubbing before the map lifts it further, most on the heavy-hub XLM-R.

				top-k var (%)		× isotropic	
enc	pair	erank	PR	$k5$	$k10$	$k5$	$k10$
mBERT	en–es	403	194	9.7	14.1	14.9	10.8
	en–zh	413	235	8.4	12.9	12.9	9.9
	en–de	408	202	9.8	14.0	15.1	10.7
	en–fr	396	183	10.4	14.9	15.9	11.4
	en–it	406	195	9.7	13.9	14.9	10.7
XLM-R	en–es	290	90	18.2	23.9	27.9	18.4
	en–zh	319	117	16.0	21.4	24.6	16.5
	en–de	300	107	16.7	23.5	25.7	18.1
	en–fr	288	92	18.3	24.4	28.2	18.7
	en–it	302	101	16.4	22.6	25.2	17.4

Table 10: Spectrum of the concatenated bilingual activation matrix (layer 1, M=2000, 5 seeds, d=768). Effective rank (erank) and participation ratio (PR) are both in the hundreds, so the space is high-dimensional; what concentrates is the top-k variance, which runs several times its isotropic share k/d, more so on XLM-R. This concentration, not a small effective rank, is what the ablation acts on.

E Layer-wise sweep, full numbers

Word-level BLI P@1 and in-degree skew at each layer, identity retrieval, M=1500, 3 seeds. mBERT word-level signal peaks at an early layer with mild skew; XLM-R is far more hub-dominated at every layer and its embedding layer (L0) is both best for word-BLI and most skewed. We report the causal analysis at layer 1 rather than the mBERT P@1 peak (layer 4) because the two sit within about a point of each other on word-level BLI, layer 1 is the layer the lineage and amplification runs share, and every causal comparison is within-representation at a fixed layer, so the choice shifts effect sizes, not the orderings.

F The hubness lineage, full numbers

Layer 1, M=2000, 5 seeds. Methods: cosine, CSLS (k=10 neighbors), inverted softmax (ISF, β=30), and a column-mean global correction in the spirit of globally-corrected retrieval (GC); note Dinu et al. (2015) define GC as rank-based retrieval, so our column-mean variant is a simplification and its weakness below speaks to that variant, not to their method. The retrieval fixes flatten the in-degree distribution and lift P@1; the inverted softmax flattens strongly when hubness is severe (XLM-R) and unreliably when mild (mBERT, β untuned); the column-mean correction is weak. On the hyper-parameters: β=30 is the value tuned on XLM-R, and we deliberately left it unretuned for mBERT rather than fitting a separate β to reach parity. mBERT's hubness is mild enough that the correction is small either way (the inverted softmax leaves mBERT skew near baseline while sharply flattening XLM-R), so an untuned mBERT baseline understates rather than inflates the retrieval fixes, and the comparison is conservative for our argument. CSLS at k=10 is the standard setting and is not sensitive in this regime. The "subspace

encoder	pair	k=1	k=2	k=3	k=5	k=10	full
mBERT	en–es	0.066	0.159	0.247	0.430	0.782	0.996
	en–fr	0.076	0.243	0.399	0.614	0.896	0.997
	en–de	0.135	0.312	0.420	0.621	0.913	0.997
	en–it	0.056	0.155	0.257	0.442	0.826	0.992
	en–bg	0.055	0.167	0.273	0.478	0.799	0.990
	en–ru	0.084	0.212	0.322	0.546	0.875	0.990
	en–fi	0.080	0.159	0.263	0.449	0.743	0.972
	en–tr	0.069	0.124	0.169	0.313	0.639	0.971
	en–ko	0.042	0.089	0.174	0.332	0.618	0.959
	en–zh	0.059	0.159	0.249	0.416	0.761	0.985
	en–ar	0.023	0.082	0.183	0.390	0.758	0.959
XLM-R	en–es	0.055	0.097	0.166	0.372	0.754	0.996
	en–fr	0.059	0.139	0.278	0.546	0.879	0.999
	en–de	0.097	0.173	0.273	0.480	0.893	0.999
	en–it	0.049	0.115	0.181	0.376	0.781	1.000
	en–bg	0.031	0.101	0.190	0.489	0.860	0.994
	en–ru	0.052	0.110	0.230	0.506	0.860	0.987
	en–fi	0.073	0.111	0.170	0.428	0.836	0.989
	en–tr	0.055	0.087	0.124	0.281	0.687	0.999
	en–ko	0.020	0.066	0.117	0.270	0.709	0.985
	en–zh	0.023	0.067	0.128	0.330	0.750	0.978
	en–ar	0.007	0.042	0.119	0.292	0.704	0.957

Table 11: Coherent-prefix bitext P@1 against input length, eleven languages, two encoders. Single deterministic pass over the fixed 712-sentence test block (no seed variation).

pair	k	coherent	shuffled	random-bag
en–es	3	0.247	0.178	0.073
en–es	5	0.430	0.261	0.101
en–es	full	0.996	0.768	n/a
en–zh	3	0.167	0.110	0.008
en–zh	5	0.365	0.243	0.010
en–zh	full	0.989	0.823	n/a

Table 12: Distinctiveness confound (mBERT L8). Random-bags stay near 0 at matched length; shuffled recovers most of the gain; coherent order supplies the rest. "n/a" marks k beyond the random-bag cap ($k \leq 8$ by construction). Within-language confusability at k=3: coherent 0.54, shuffled 0.49, random-bag 0.60.

tie" columns are the Pearson correlation, across the M=2000 gallery targets, between (i) baseline in-degree and projection onto the top-10 PC subspace, and (ii) the CSLS in-degree reduction and that same projection. They are positive but modest: the hub mass CSLS corrects leans into the localized subspace above chance, not exclusively. The redundancy columns give in-degree skew under base / CSLS / ABTT / ABTT+CSLS: ABTT alone reproduces much of CSLS's flattening (on XLM-R en–zh, more than CSLS), and the two are only partly additive.

G Per-word difficulty breakdowns

mBERT layer 1, identity retrieval. Polysemy, anti-hub target status, and subword fertility each predict per-word failure univariately (Table 16). Polysemy is operationalised as the MUSE dictionary translation count, which is dense and reliable only where the dictionary is large (en–es). For en–zh the translation lists are sparse, so the polysemy and POS contrasts are marked n/a (not measured) rather than estimated from an unreliable count; a sense inventory such as BabelNet could give a script-independent polysemy measure and is left to future work.

Multivariate control. The univariate contrasts confound each other: rarer words are split into more subwords, longer words are rarer, and so on. A single logistic regression of per-word success on all five standardised factors at once (Table 15, en–es, N=4000, overall P@1 0.588) separates them. Polysemy is the strongest predictor and the only large coefficient. A function-word indicator and frequency-rank also survive (both modest; rarer words slightly harder), which sharpens the body's earlier univariate reading of frequency as null: under full-vocabulary identity retrieval there is a small but reliable frequency effect, well below polysemy.

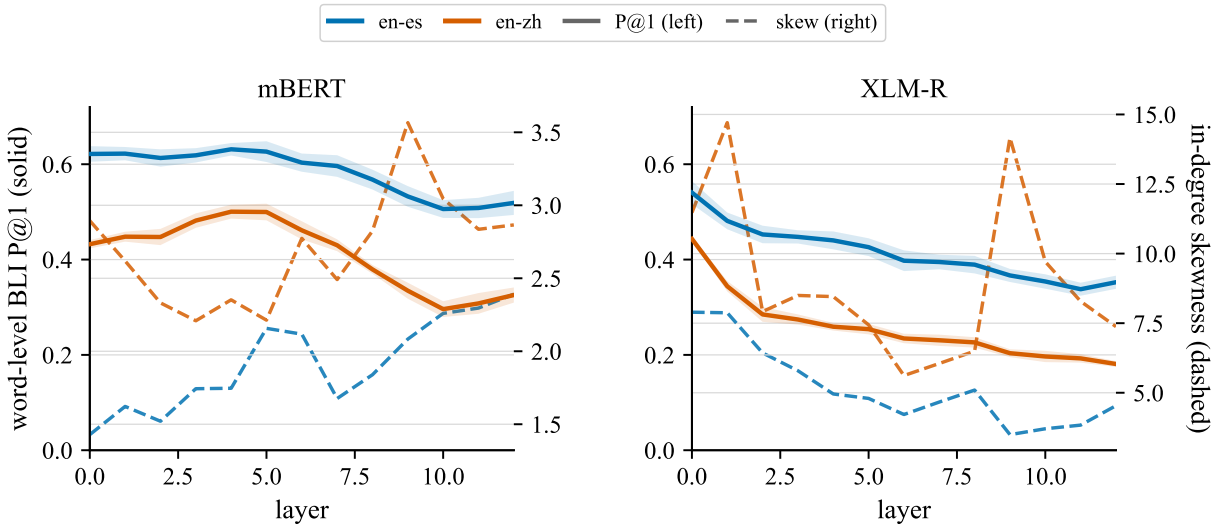


Figure 6: **Layer-wise word-level BLI P@1 (solid, left axis) and in-degree skewness (dashed, right axis).** On mBERT the word-level signal peaks at an early layer where hubness is mild and falls as hubness grows with depth. XLM-R is far more hub-dominated at every layer (note the larger skew range); its embedding layer is best for word-level BLI but also the most skewed, so the cross-layer relation is representation-dependent. The robust mechanism is within-representation, not a universal depth law.

enc/pair	metric	L0	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	L11	L12
mBERT es	P@1	.62	.62	.61	.62	**.63**	.63	.60	.60	.57	.53	.51	.51	.52
mBERT es	skew	1.4	1.6	1.5	1.7	1.7	2.2	2.1	1.7	1.8	2.1	2.3	2.3	2.4
mBERT zh	P@1	.43	.45	.45	.48	**.50**	.50	.46	.43	.38	.33	.30	.31	.33
mBERT zh	skew	2.9	2.6	2.3	2.2	2.4	2.2	2.8	2.5	2.8	3.6	3.0	2.8	2.9
XLM-R es	P@1	**.54**	.48	.45	.45	.44	.43	.40	.39	.39	.37	.35	.34	.35
XLM-R es	skew	7.9	7.9	6.4	5.8	5.0	4.8	4.2	4.7	5.1	3.5	3.7	3.8	4.5
XLM-R zh	P@1	**.44**	.34	.28	.27	.26	.25	.23	.23	.23	.20	.20	.19	.18
XLM-R zh	skew	11.5	14.7	7.9	8.5	8.5	7.4	5.6	6.1	6.5	14.2	9.7	8.3	7.4

Table 13: Layer-wise word-level BLI P@1 and in-degree skew. Bold marks the best P@1 layer.

Subword fertility and word length do not survive: their univariate signal is absorbed once polysemy and length are controlled. Coefficients are standardised log-odds with 95% bootstrap CIs over 2000 resamples of the words.

H Hubness-accuracy covariation and a per-word indirect effect

On mBERT en–es, plain cosine retrieval gives P@1 0.320, peak in-degree 39, and 55% never retrieved; CSLS gives 0.372, peak in-degree 12, and 44% never retrieved. The standard hubness correction both raises accuracy and flattens the in-degree distribution, the same quantity the directed ablation acts on. Across context lengths, hubness (in-degree skewness) and P@1 co-vary at $r = -0.72$. This is a covariation, not a fitted mediation of the length curve; the causal weight of the paper is the direct intervention in §6.

A per-word indirect effect (null). We also tested the first step of the chain per word, which does not need a gallery-level model. On the en–es per-word data (mBERT layer 1, N=4000; Table 15) the treatment is log polysemy, the mediator is each word's projection onto the top-10 hub subspace (the geometry ABTT removes), and the outcome is retrieval success. The indirect effect is a product-of-coefficients indirect effect of -0.001 with a BCa 95% interval of $[-0.011, +0.009]$ through the hub-subspace projection, and -0.005 with $[-0.021, +0.012]$ through a mean-cosine centrality mediator, both over B=2000 bootstrap resamples of the words. Both intervals include zero, so the per-word indirect effect is not individually significant. We report this rather than bury it: polysemy remains the strongest per-word predictor of failure jointly (Table 15), but the specific path through this hub-subspace mediator is not resolved at the word level, so the paper's causal weight is the intervention (§6) and at the length level we claim covariation, not mediation.

enc	pair	P@1 cos/CSLS/ISF/GC	skew cos/CSLS/ISF	tie N_1/ΔCSLS	skew b/C/A/A+C
mBERT	en–es	.609/.636/.664/.607	1.74/0.92/2.45	0.16/0.23	1.74/0.92/1.19/0.76
	en–zh	.419/.472/.493/.417	2.46/0.97/0.92	0.34/0.35	2.46/0.97/2.23/1.00
	en–de	.418/.448/.460/.416	2.44/1.40/2.17	0.27/0.25	2.44/1.40/2.01/1.14
XLM-R	en–es	.479/.499/.523/.477	9.68/7.53/2.14	0.17/0.29	9.68/7.53/10.02/6.22
	en–zh	.342/.372/.397/.337	15.76/9.36/1.92	0.40/0.36	15.76/9.36/5.47/4.54
	en–de	.367/.387/.395/.365	15.32/9.23/3.71	0.16/0.21	15.32/9.23/9.64/5.95

Table 14: Lineage retrieval fixes vs the localized subspace (layer 1, 5 seeds; 5-seed means, CIs omitted). ISF is the inverted softmax; GC the column-mean correction; b/C/A/A+C is base/CSLS/ABTT/ABTT+CSLS.

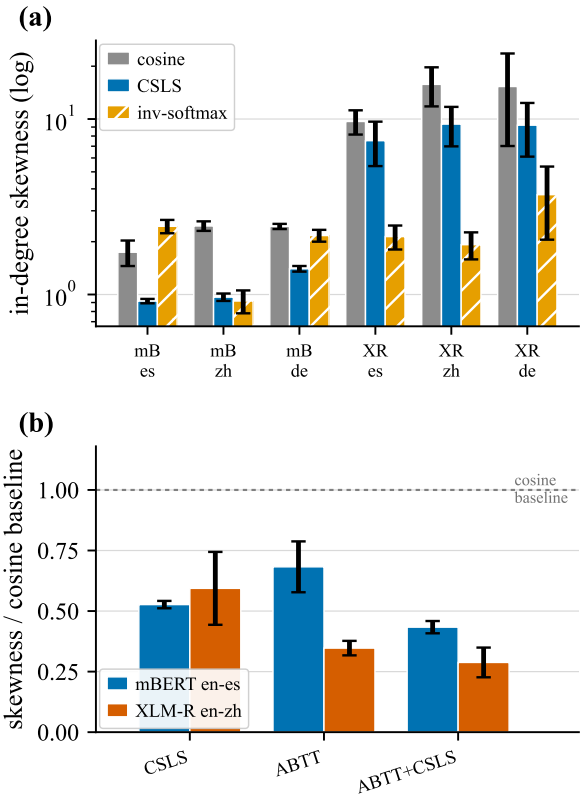


Figure 7: **Retrieval-side hubness fixes and the localized subspace. (a)** In-degree skewness (log) under cosine, CSLS and inverted softmax across model×pair cells. The retrieval fixes flatten the distribution. **(b)** Skewness as a fraction of the cosine baseline for two cells.

factor	std. coef. (95% CI)	survives
log polysemy	-0.60 $[-0.67, -0.53]$	yes
log freq-rank	-0.32 $[-0.40, -0.23]$	yes
function word	-0.20 $[-0.28, -0.12]$	yes
log fertility	$+0.01$ $[-0.07, +0.08]$	no
log length	-0.01 $[-0.08, +0.05]$	no

Table 15: Multivariate logistic regression of en–es per-word retrieval success on all five factors jointly (mBERT layer 1, identity retrieval, N=4000 words). Standardised coefficients; "survives" marks a significant coefficient, i.e. a 95% bootstrap CI excluding 0. Polysemy dominates; fertility and length wash out under control.

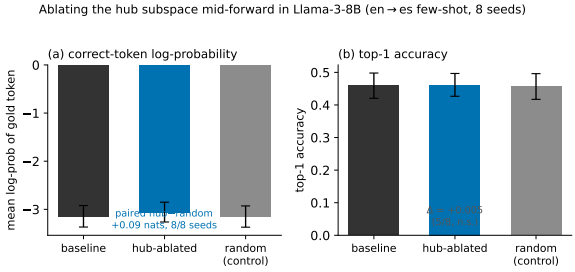


Figure 8: Mid-forward ablation of the hub subspace in Llama-3-8B. The task is en→es few-shot translation with 8 seeds and N=150. **(a)** The ablation raises the log-probability of the correct token. **(b)** Top-1 accuracy does not change. Random ablation stays at baseline.

I Secondary results

Neighborhood preservation. The mean over k=10 of within-language k-NN overlap between a word and its translation tracks BLI across pairs and architectures (r=0.96 across eight model×pair cells) and per-word retrieval on en–es (r=0.54, N=4000 words). It is centering-invariant, so it is a cross-pair difficulty predictor rather than a within-pair lever.

Decoder mid-forward ablation, per-condition. Few-shot en→es translation by Llama-3-8B, 8 seeds, N=150 (unit-normalized hub subspace). Correct-translation log-prob: baseline $-3.144\pm.269$, hub-ablation $-3.057\pm.248$, random $-3.150\pm.266$. Top-1 accuracy: 0.459 / 0.462 / 0.457. Top-5 accuracy: 0.700 / 0.707 / 0.700. Gold-token median rank: 1.9 / 1.9 / 1.9. Paired hub−random deltas: $\Delta \log p = +0.093\pm.050$ (positive in 8/8 seeds), ΔMRR $= +0.007\pm.008$ (6/8), Δtop-1 $= +0.005\pm.013$ (5/8), Δtop-5 $= +0.007\pm.012$. The reliable effect is on the correct token's log-probability; accuracy and rank do not move at this sample size. Ablating the *raw* (non-normalized) subspace instead collapses generation (top-1 0.02), an outlier-dimension artifact we discuss in Appendix N.

factor	contrast	en–es	en–zh
polysemy	P@1 monosemous	0.517	n/a
polysemy	P@1 $\geq$3 translations	0.259	n/a
anti-hub	P@1 target N_1=0	0.023	0.000
anti-hub	P@1 target $N_1 \geq 10$	0.649	0.154
morphology	P@1 subwords $\leq$1	0.433	0.112
morphology	P@1 subwords $\geq$3	0.237	0.077
frequency	corr(freq-rank, correct)	~ 0.01	-0.04
POS	P@1 function words	0.150	n/a
POS	P@1 content words	0.203	n/a

Table 16: Per-word difficulty. "n/a" marks cells not measured for that pair.

The interpretable-axis negative. PC0 correlates with subword fertility (R^2=0.58 en–es, 0.29 en–zh), but the dedicated fertility and length axes are nearly orthogonal to the causal hub subspace (principal angle 81.3° en–es, 86.1° en–zh), and ablating them behaves like random ablation (0.121 / 0.119 vs baseline 0.118), not like top-PC ablation (0.177). The hub subspace overlaps tokenization structure in part but is not reducible to it.

The language-identity negative. Per-language centering removes not just the language means but linear language separability: a held-out logistic probe on the stacked centered space classifies en vs. target at only 0.38–0.46 accuracy across the ten model×pair cells, at or below chance. The direction it fits is noise-like: it sits 83.1° from the hub subspace on average, the angle a random direction makes with a ten-dimensional subspace. Ablating it moves anti-hub recovery like the one-random-direction control (0.082 vs. 0.081 mean) while ABTT-k1 reaches 0.128 (per-cell numbers released with the code). The dial is not a language-ID axis, matching the fertility and length negatives above; Appendix J sets this negative against the language-identity-removal family.

J Relation to language-identity subspace removal

A line of work from 2020 onward projects a small subspace out of multilingual encoder representations and improves cross-lingual retrieval, reading the removed structure as language identity. Libovický et al. (2020) subtract per-language centroids and sharply improve parallel-sentence retrieval; LIR (Yang et al., 2021) finds that the principal components of multilingual sentence spaces primarily encode language identity and removes them by orthogonal projection, with large gains on cross-lingual QA retrieval; LSAR (Xie et al., 2022) fits a low-rank subspace by SVD over monolingual corpora and projects embeddings into its null space with no finetuning; Hämmerl et al. (2023) reach similar improvements by removing outlier dimensions or applying ZCA whitening; the line is still active (Kim et al., 2026). Operationally, these methods and our ABTT step look alike: estimate a small subspace without supervision, project it out, and cross-lingual retrieval improves.

Three points separate our subspace from theirs. First, our pipeline applies per-language centering before any subspace is estimated, and centering already removes the linear language-identity structure this family targets. The held-out language probe of Appendix I classifies en vs. target at 0.38–0.46, at or below chance; the fitted language direction sits 83.1° from the hub subspace; and ablating that direction behaves like a random control. The hub subspace is what remains causally active after the language-identity signal is gone, so it is not the structure these methods remove. Second, none of these works measure hubness: no in-degree distributions, no skewness, no anti-hubs. They also intervene in one direction only, which establishes that a projection helps, not what the removed structure does; the amplification arm and the matched random-direction controls (§6, App. M) are what upgrade a helpful projection to a causal attribution. Third, the family operates on sentence or passage representations, while our effect is at the word level, where hubness is most severe (§4), with recovery of anti-hub words as the readout. The whitening comparison of Appendix B.3 is the point of closest contact: Hämmerl et al. (2023) also apply ZCA whitening, at the sentence level; we use it as convergent evidence that the de-hubbing lives in the top-variance directions.

The mechanism has a theoretical ancestry worth naming. That spatially central points become hubs, and that centering-type corrections reduce them, is established for general similarity spaces: Suzuki et al. (2013) show that centering a similarity mea-

sure reduces hubs, and Hara et al. (2016) trace hubness to the density gradient around the data centroid. The account in §5, ambiguous words landing in a dense central region and winning the nearest-neighbor competition, instantiates that theory in a multilingual model's activation space. What is new is the localization of the hub-carrying variance to identified directions of a learned representation, and the bidirectional intervention on those directions, rather than a correction applied to the similarity measure or metric.

K Negative results

Relative representations beat no fair baseline. Under multi-seed evaluation in a full-vocabulary regime (diverse anchors, full-vocabulary centering, no PCA, 8 seeds), a tuned unsupervised self-learner (Artetxe et al., 2018) reaches 0.935 (en–es) and 0.954 (en–zh) from zero gold anchors; the relative-representation hybrid never exceeds it. A no-map **identity** retrieval is itself a strong baseline (en–es 0.688, en–zh 0.576, above single-shot Procrustes (Smith et al., 2017) at 0.626 and 0.290). The apparent distant-pair gap in a capped regime (en–zh unsup 0.363) was a regime artifact; it closed to 0.954 with a diverse unlabeled vocabulary and full centering.

The typological floor. On genuinely low-resource, different-script pairs (PanLex fi–ko, ko–tr), every method sits near zero; an XLM-R control replicated the near-zero result, though those exact values were not persisted to the run log.

L Sentence-level XNLI probe

Our account makes a falsifiable prediction at the sentence level. Because context de-hubs the representation (§4, §5), single-token hubness should stop mattering once inputs are sentences, *except* on an encoder whose hubness is severe enough to survive contextual averaging. So removing the hub subspace should help zero-shot sentence transfer on the strongly hub-dominated XLM-R but not on the mildly-hub mBERT, and single-token hubness should not predict the per-language transfer gap. Both hold (Fig. 9; Table 17). The effect is real but modest and one-sided, and the paper's headline causal result is at the word level.

Setup. Frozen cross-encoder features (premise and hypothesis encoded together, masked-mean pooled at the MNLI-dev-selected layer, L9 for both encoders), a linear head on 25k English MNLI, 5 seeds; zero-shot XNLI test (Conneau et al., 2018, ∼5010 examples/language). "acc" is mean accuracy, "gap" is en-acc minus target-acc; "skew$_1$" and "P@1$_1$" are per-language single-token hubness measures from FLORES. The transfer gap shows no detectable correlation with single-token hubness across the n=14 target languages (gap vs. skew: mBERT Pearson r=0.11, p=.72, Spearman ρ=0.23, p=.43; XLM-R r=0.22, p=.45, ρ=0.16, p=.59; gap vs. single-token P@1: $r=-0.20$, p=.49 and $r=-0.04$, p=.90), consistent with context already de-hubbing sentence representations. With 14 languages this test only rules out strong correlations ($|r| \gtrsim 0.5$ at $p<.05$), so we read it as an absence of evidence for a gap–hubness link, not a demonstrated null.

Causal ablation (mean target-language accuracy, 5 seeds, 95% CI), projecting out the top-10 PC hub subspace vs an equal-rank random-direction control: mBERT baseline 0.471±.003, random 0.472±.003, hub 0.455±.002 (paired hub−random −0.017±.003, 0/5 seeds positive); XLM-R baseline 0.431±.005, random 0.432±.006, hub 0.448±.001 (paired +0.016±.007, 5/5). English accuracy moves by ≤ 0.005 in all conditions. The probe extracts cross-encoder sentence features and measures their per-language single-token hubness, then runs the causal ablation against the random-direction control.

M Amplifying the hub subspace

The ablation in §6 (App. B) shows the hub subspace is necessary to relieve cross-lingual hubness. The symmetric test is to scale it *up*. We map each centered activation $a \mapsto a + (\alpha - 1)(aQ^\top)Q$ for Q an orthonormal basis of the top-K=10 subspace (the same subspace the ablation removes), then renormalise; α=1 is baseline and projecting-out is the α=0 limit. As a specificity control we amplify K random orthonormal directions by the same α. All conditions are read off the same cached layer-1 activations over 20 seeds (Fig. 10, Table 18).

Amplifying the hub subspace collapses cross-lingual retrieval on all ten cells: P@1 falls monotonically with α (absolute drop 0.19 to 0.41 from baseline to α=8) and anti-hub recovery falls toward zero ($\leq$ 0.04 at α=8), while the never-retrieved fraction rises. Hub-ablation moves the same quantities the opposite way (Table 1). Amplifying random directions of equal rank leaves P@1 within

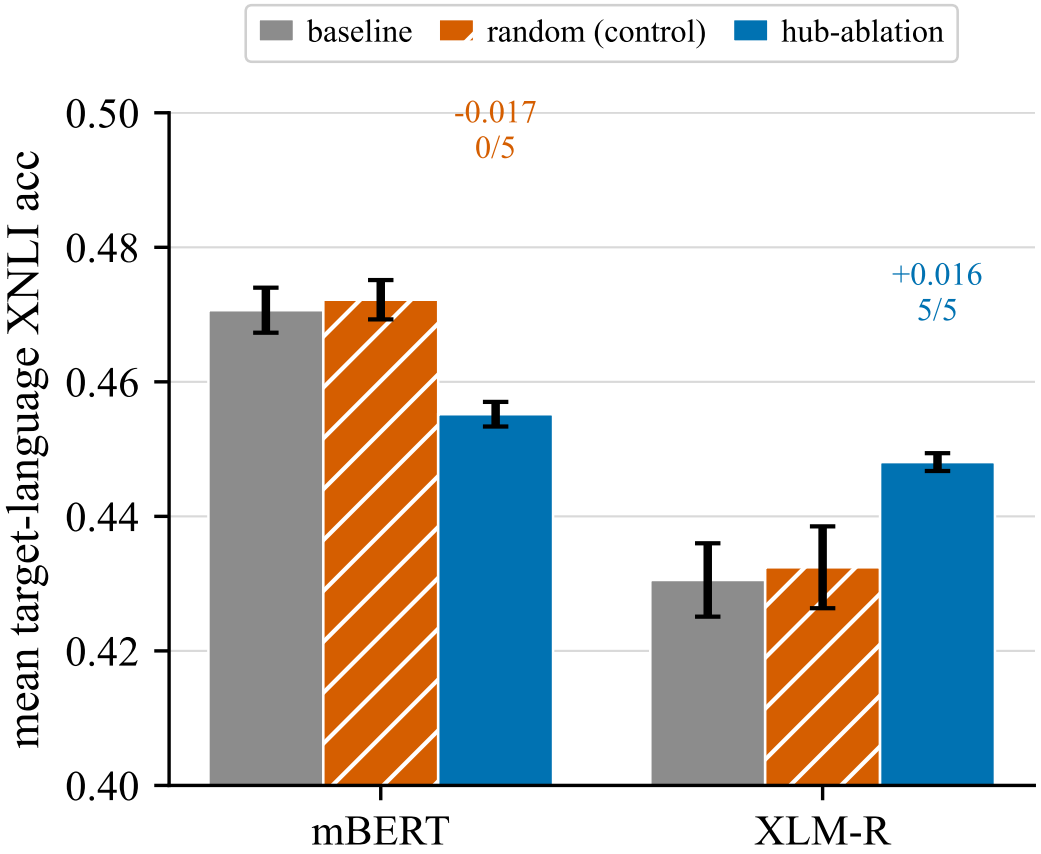


Figure 9: **Sentence-level causal test (zero-shot XNLI).** Mean target-language accuracy under baseline, random-direction ablation, and hub-subspace ablation (5 seeds, 95% CIs). Removing the hub subspace beats the random control on the hub-dominated XLM-R ($+0.016$, $5/5$ seeds) but not on mBERT (-0.017, $0/5$), the contrast our theory predicts.

noise of baseline on eight of ten cells; on en–zh, the lowest-baseline pair, it lowers P@1 modestly (mBERT 0.42 $\rightarrow$ 0.34, XLM-R 0.34 $\rightarrow$ 0.29), still several times smaller than the hub-subspace collapse on that pair (to 0.06 and 0.03).

The form of the induced damage depends on the regime. On mBERT, where baseline single-token hubness is mild (max in-degree 9–14), amplification concentrates in-degree into larger hubs on all five pairs (e.g. en–de 13 $\rightarrow$ 63). On XLM-R, already a heavy-hub regime at layer 1 (baseline max in-degree 33–75), the same scaling disperses in-degree on four of five pairs (e.g. en–de 67 $\rightarrow$ 39) and leaves it flat on en–zh, even as retrieval collapses and %never rises. Once the hub subspace already dominates the geometry, amplifying it strands words broadly rather than through a single dominant hub, which matches the §6 observation that XLM-R is the heavier-hub regime where ablation exceeds CSLS. The sufficiency claim is therefore on retrieval collapse and stranding, the behavioral pathology, not on peak in-degree.

N Reproducibility

Activations. Words are embedded in isolation, one word per forward pass, subword tokens mean-pooled, then per-language mean-centered. We report mBERT (`bert-base-multilingual-cased`) and XLM-R_{base} (`xlm-roberta-base`) layer 1 for word-level retrieval and layer 8 masked-mean for sentence-level; Llama-3-8B at 4-bit (MLX) layer 16, last-token, for the decoder. **Data.** MUSE dictionaries for en–{es,de,fr,it,zh}; FLORES-200 for parallel sentences; PanLex for the distant-pair floor. **Retrieval.** Identity (raw cosine) NN for the causal analyses; CSLS with k=10, inverted softmax (β=30), and the column-mean GC variant (App. F) as the lineage comparisons. **Hub directions** are the top principal components of the concatenated, per-language-centered bilingual activation matrix (source and target activations stacked, then the joint top PCs taken by SVD); ablation projects them out of both source and target. Subspace stability across seeds, the

lang	mBERT acc	gap	XLM-R acc	gap	skew$_1$	P@1$_1$
en	0.569	–	0.534	–	–	–
ar	0.450	0.119	0.419	0.115	21.5	0.018
bg	0.513	0.056	0.403	0.131	20.9	0.044
de	0.438	0.131	0.379	0.155	23.3	0.114
el	0.470	0.099	0.435	0.099	19.0	0.016
es	0.525	0.043	0.453	0.082	19.7	0.063
fr	0.515	0.053	0.465	0.069	20.7	0.073
hi	0.443	0.126	0.371	0.164	20.7	0.032
ru	0.501	0.068	0.458	0.076	23.0	0.071
sw	0.439	0.130	0.412	0.122	18.9	0.065
th	0.386	0.183	0.449	0.085	20.5	0.028
tr	0.443	0.126	0.469	0.065	21.2	0.058
ur	0.432	0.137	0.362	0.172	22.0	0.033
vi	0.521	0.048	0.476	0.058	19.4	0.064
zh	0.514	0.055	0.477	0.057	21.9	0.033

Table 17: Zero-shot XNLI accuracy and transfer gap per language (5-seed means), with per-language single-token hubness.

encoder	pair	P@1 ABTT	P@1 base	P@1 amp×8	P@1 rnd×4	max N_1 b→amp	%never b→amp
mBERT	en–es	0.627	0.607	**0.205**	0.585	11 → 27	36 → 60
	en–zh	0.414	0.420	**0.058**	0.336	14 → 98	47 → 62
	en–de	0.441	0.420	**0.105**	0.400	13 → 63	51 → 66
	en–fr	0.623	0.599	**0.189**	0.579	9 → 31	37 → 61
	en–it	0.590	0.567	**0.181**	0.548	11 → 30	39 → 62
XLM-R	en–es	0.542	0.474	**0.223**	0.457	37 → 20	42 → 51
	en–zh	0.439	0.339	**0.034**	0.292	75 → 78	53 → 70
	en–de	0.448	0.372	**0.184**	0.354	67 → 39	52 → 57
	en–fr	0.539	0.473	**0.266**	0.456	33 → 20	43 → 49
	en–it	0.536	0.468	**0.241**	0.452	47 → 33	44 → 53

Table 18: Bidirectional dial (layer 1, K=10, 20 seeds; 20-seed means, CIs omitted). Bold marks the amplified condition (the collapsed value, not the best). Amplifying the hub subspace (amp×8) collapses P@1 far below baseline on every cell, while equal-rank random amplification (rnd×4) stays near baseline and hub-ablation (ABTT) sits above it. Max in-degree (max N_1) rises under amplification on mBERT and falls on the hub-saturated XLM-R; %never rises on both. "b" is baseline, "amp" is α=8.

k-sweep, the whitening comparison, and the multivariate per-word regression are all computed on these cached layer-1 activations, with no extra model forward passes. In the headline tables the subspace is estimated on the same M=2000 draw it is applied to, the standard ABTT protocol. Appendix B.4 repeats the causal comparison with strict separation, subspace and centering fit on a disjoint train half or on a different language pair, via a deterministic script released with our code. **Decoder hub subspace.** For the Llama-3-8B mid-forward ablation the hub directions are estimated from the decoder's own activations, not transferred from the encoder PCs. We stack Llama's layer-16 last-token vectors for the deduplicated en and es MUSE words, *unit-normalize* each vector, subtract a single global mean over the stacked matrix, and take the top ten right singular vectors. Unit-normalizing is necessary: without it the top components of raw Llama activations are the few massive activations of the decoder (Sun et al., 2024), and projecting those out collapses generation (top-1 accuracy falls from 0.46 to 0.02), which is a property of those dimensions, not of the hub structure. Normalizing, as the encoder analyses do, isolates the hub subspace. The subspace is bilingual, fit jointly over the two languages rather than for the en→es direction; we orthonormalize it and project it out of the running activations at layer 16, against an equal-rank random-direction control. **Seeds and samples.** Ablation and lineage tables use M=2000 and 5 seeds; the layer sweep M=1500 and 3 seeds; the decoder mid-forward ablation 8 seeds, N=150, 5 in-context examples; App. A states what each protocol's seed controls. **Code.** Code and configuration files to reproduce all experiments and figures are included in the supplementary material and will be released publicly upon acceptance.

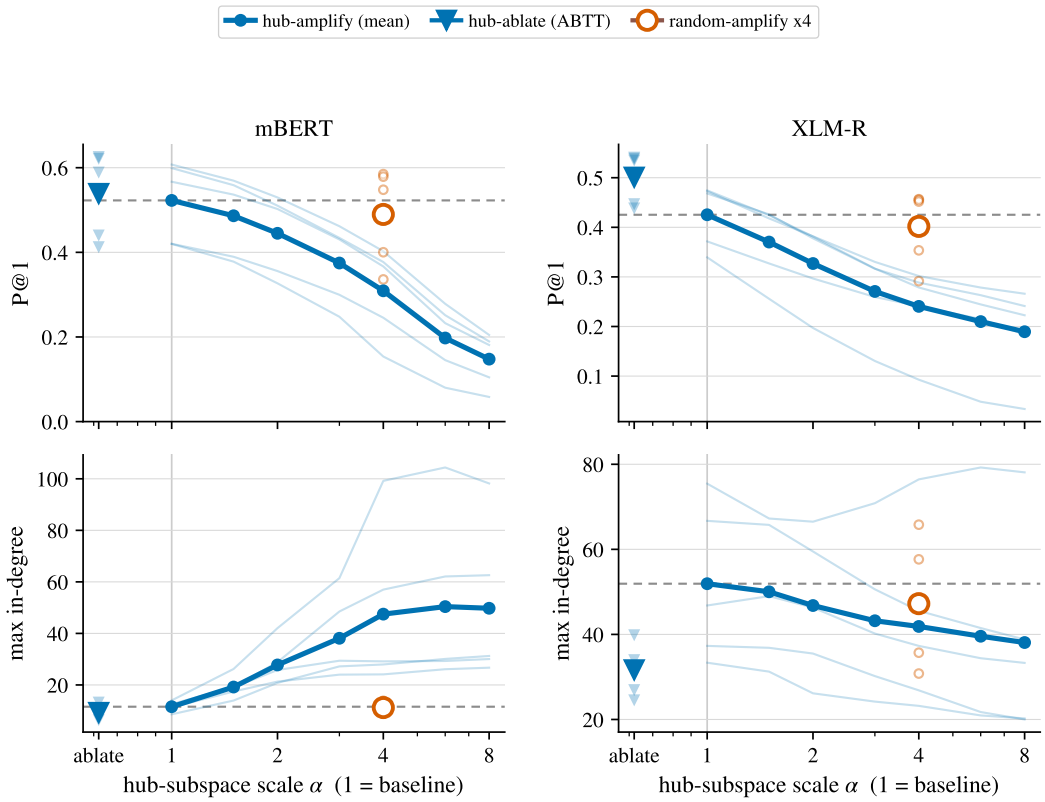


Figure 10: Turning the hub subspace down (ablate, ABTT) restores cross-lingual retrieval; turning it up (amplify, $\alpha > 1$) collapses it, on both encoders (top row). Bold is the mean over five pairs, faint lines are per pair; random-amplify $\times 4$ (open marker) is the specificity control. In-degree concentration (bottom row) rises for mBERT but disperses for the already hub-saturated XLM-R; the retrieval collapse is common to both. Layer 1, $K{=}10$, 20 seeds.